Karlsruhe Institute of Technology

Intelligent Sensor-Actuator-Systems

Variational Optimization with Deterministic Sampling for Model Predictive Control

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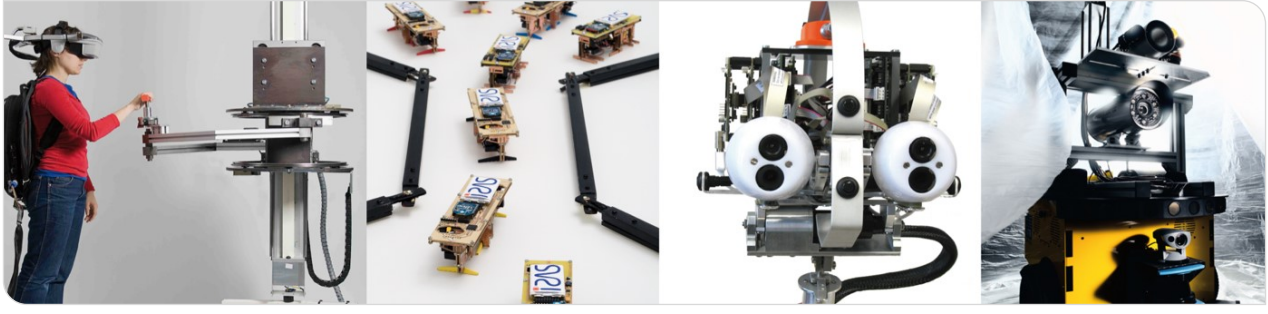
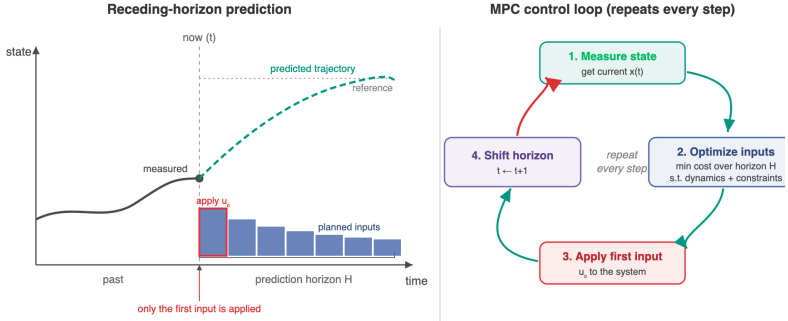


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Motivation: Model Predictive Control

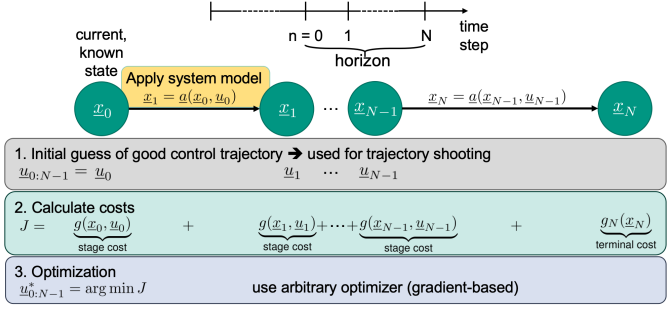
- Predict → Optimize → Apply first input → Repeat
- Handles nonlinear dynamics and constraints



Predict the future, optimize inputs over the horizon, apply only the first, then re-measure and repeat.

Motivation: Model Predictive Control

- At each step: find the input sequence $U = (u_0, \ldots, u_{H-1})$ that minimizes the predicted cost $J(U)$
- Why it's hard:
  - Non-convex $\rightarrow$ prone to local minima
  - High-dimensional $\rightarrow$ scales with horizon $\times$ inputs
  - Time-critical $\rightarrow$ re-solved at every control step



Blocks

Contentblock

Colourless block with bold heading.

Greenblock

Default (`block`)

Blueblock

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Redblock

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Brownblock

Purpleblock

Cyanblock

Yellowblock

Lightgreenblock

Orangeblock

Grayblock

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Example Content

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Example Content: Literature

Citation: Klare et al. 2021

Literature

Backup Section

Slides inserted after `\beginbackup` do not count towards the total number of slides.

[1] Heiko Klare et al. “Enabling consistency in view-based system development – The Vitruvius approach”. In: *Journal of Systems and Software* 171 (2021). ISSN: 0164-1212. DOI: https://doi.org/10.1016/j.jss.2020.110815. URL: http://www.sciencedirect.com/science/article/pii/S0164121220302144.

Colour Palette

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